

Name : P. Sonu

Reg No: 192365057

Course Code: CSA1007

Agile Sprint Planning & User Story Workshop

Smart Public Library Management Platform

1. Project Vision and Stakeholders

Project Vision

The vision of the **Smart Public Library Management Platform** is to develop a distributed, cloud-based library management system that automates book circulation, digital resource management, membership activities, reservations, inventory tracking, and analytics across multiple library branches. The platform aims to provide real-time book availability, personalized recommendations, automated notifications, demand prediction, and efficient collection management, thereby improving operational efficiency and user satisfaction.

Project Objectives

- Automate book issue, return, and reservation processes.
- Track books and digital resources in real time.
- Improve inventory management across multiple branches.
- Recommend books based on user interests.
- Predict demand for popular books.
- Send automatic due-date and reservation notifications.
- Generate reports and analytics for administrators.
- Improve patron satisfaction and reduce overdue returns.

Key Stakeholders

Stakeholder	Roles and Responsibilities
Library Members	Search books, reserve books, borrow/return books, access digital resources, receive notifications.
Librarians	Manage books, memberships, issue/return books, maintain inventory, assist users.
Library Administrator	Manage branches, monitor reports, analyze book demand, manage staff.
System Administrator	Maintain servers, databases, security, backups, and system availability.
Government/Library Authority	Monitor library performance, approve policies and funding.
Book Suppliers	Supply new books according to demand predictions.

2. Epics and User Stories

Epic 1: User and Membership Management

User Story 1

As a library member, **I want** to register an account, **so that** I can access library services online.

User Story 2

As a librarian, **I want** to approve or reject membership requests, **so that** only eligible users become members.

Epic 2: Book Management

User Story 3

As a librarian, **I want** to add, edit, and delete book records, **so that** the library catalog remains updated.

User Story 4

As a library member, **I want** to search books by title, author, or category, **so that** I can easily find books.

Epic 3: Book Circulation

User Story 5

As a librarian, **I want** to issue and return books electronically, **so that** transactions are completed quickly.

User Story 6

As a member, **I want** to renew borrowed books online, **so that** I can extend the borrowing period.

Epic 4: Reservation System

User Story 7

As a member, **I want** to reserve unavailable books, **so that** I can borrow them when available.

Epic 5: Notifications

User Story 8

As a member, **I want** to receive reminders before the due date, **so that** I can avoid overdue fines.

Epic 6: Recommendation and Analytics

User Story 9

As a member, I want personalized book recommendations, **so that** I can discover books matching my interests.

User Story 10

As a library administrator, **I want** reports on book circulation and demand, **so that** I can improve collection planning.

3. Acceptance Criteria

User Story 1 – Member Registration

Given a new user visits the registration page

When the user enters valid details and submits the form

Then a new member account should be created successfully.

User Story 2 – Membership Approval

Given a pending membership request

When the librarian approves it

Then the member should receive account activation.

User Story 3 – Book Management

Given the librarian is logged in

When the librarian adds a new book

Then the book should appear in the library catalog.

User Story 4 – Search Books

Given books exist in the library

When the member searches using title or author

Then matching books should be displayed.

User Story 5 – Issue and Return Books

Given a book is available

When the librarian issues the book

Then the inventory status should change to "Issued."

User Story 6 – Renew Book

Given the borrowed book has no reservation

When the member requests renewal

Then the due date should be extended.

User Story 7 – Reserve Book

Given the selected book is unavailable

When the member clicks Reserve

Then the reservation should be added to the waiting list.

User Story 8 – Due Date Notification

Given a due date is approaching

When one day remains

Then an automatic reminder should be sent.

User Story 9 – Book Recommendation

Given borrowing history exists

When the member logs in

Then recommended books should be displayed.

User Story 10 – Analytics

Given borrowing data exists

When the administrator generates reports

Then circulation statistics and demand reports should be displayed.

4. Product Backlog Prioritization

Priority	Product Backlog Item	Business Value
High	User Registration & Login	Very High
High	Book Search	Very High
High	Book Issue & Return	Very High
High	Book Management	Very High
High	Reservation System	High
Medium	Notifications	High
Medium	Digital Resource Management	Medium
Medium	Recommendation System	Medium
Low	Demand Prediction	Medium
Low	Analytics Dashboard	Medium

5. Sprint Backlog (Sprint 1)

Sprint Duration

2 Weeks

Selected User Stories

User Story	Tasks
Member Registration	Design registration page, validate inputs, create database tables, implement login
Book Search	Design search page, implement search API, display search results
Book Management	Add/Edit/Delete books, connect database
Issue & Return Books	Develop issue module, return module, update inventory automatically

6. Story Point Estimation

Estimation Technique: Fibonacci Sequence

User Story	Story Points Justification	
Member Registration	5	Includes UI, validation, database integration
Book Search	3	Moderate search functionality
Book Management	8	CRUD operations with inventory updates
Issue & Return Books	8	Business logic and database transactions
Reservation System	5	Queue handling and notifications
Notifications	3	Email/SMS integration
Recommendation System	13	Machine Learning algorithm required
Analytics Dashboard	8	Charts and report generation
Demand Prediction	13	Predictive analytics using historical data

Justification

The Fibonacci sequence (1, 2, 3, 5, 8, 13, 21) is used because it reflects increasing uncertainty and complexity. Features with higher complexity, such as recommendation systems and demand prediction, receive higher story points, while simpler features like search and notifications receive fewer points.

7. Sprint Goal and Expected Deliverables

Sprint Goal

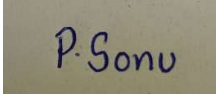
The goal of Sprint 1 is to build the foundation of the Smart Public Library Management Platform by implementing secure user registration, authentication, book catalog management, book search, and book circulation (issue and return) features. These core functionalities will allow the system to perform essential library operations and prepare the platform for future enhancements such as reservations, notifications, analytics, and AI-based recommendations.

Expected Sprint Deliverables

- User Registration and Login Module
- Member Authentication System
- Book Catalog Management Module
- Book Search Functionality
- Book Issue and Return Module
- Database Schema for Members and Books
- Basic User Interface
- Unit Testing Report
- Integration Testing Report
- Sprint Review Documentation
- Updated Product Backlog for Sprint 2

Conclusion

The Agile Sprint Planning and User Story Workshop provides a structured roadmap for developing the Smart Public Library Management Platform using Agile Scrum practices. The project begins with a well-defined vision, identifies all relevant stakeholders, and organizes requirements into epics and user stories with measurable acceptance criteria. The product backlog is prioritized according to business value, while Sprint 1 focuses on delivering the essential features required for day-to-day library operations. Story point estimation using the Fibonacci sequence supports realistic effort planning, and the sprint goal ensures that the team delivers a functional foundation that can be expanded incrementally in future sprints. This Agile approach promotes flexibility, continuous feedback, high software quality, and successful project delivery.

A small, rectangular, light brown textured box containing the handwritten text "P. Sonu" in a dark blue or purple ink.